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Quantitative decision science

M.A. Actuarial Economics - Madras School of Economics - Class of 2028
Summer 2027 internships - 9 core projects with source-linked evidence

Consumer-card strategy, US loss forecasting and bank-model validation lead the portfolio. Causal decisions, operations, finance, economics, treasury and insurance add distinct evidence. Each project exposes its data, methods, results and limitations.

1. Consumer credit risk and strategy

The boosting model's historical 600-client review queue (10% capacity) captured 33.4% of observed defaults; 72% of queued clients defaulted. A shallow CART benchmark and five cost assumptions expose segmentation and workload tradeoffs. Models use 30,000 Taiwanese clients from 2005, with independent SQL reconciliation. A separate 56-quarter US bankcard reporting pack tracks delinquency and origination mix; it is not joined to the borrower sample. Concentration does not establish prevented losses or profit.

Consumer credit, risk strategy, model validation and credit analytics
Code and research - Explore card strategy

2. US card loss forecasting and scenarios

Archived ALFRED inputs support 46 quarterly US aggregate card-loss evaluations, with train-only scaling and an embargoed model-selection snapshot. Empirical error bands and a formula workbook make uncertainty and hypothetical exposure accounting inspectable. The AR model's small gain is uncertain; the macro model fails during COVID. These are aggregate research forecasts and scenarios, not borrower default probabilities or CECL reserves.

Credit-loss forecasting, model risk, risk analytics and financial reporting
Code and research - Research report

3. Bank-risk model validation

237,253 annual bank observations support a purged temporal failure-monitoring study with separate calibration and policy stages. Later monitoring missed 9 of 10 failures, restricting use despite stronger earlier ranking. Bank failure is not borrower default or regulatory approval.

Bank risk, model validation, risk analytics
Code and research - Evidence browser

4. Causal customer decisions

A 64,000-customer randomized campaign did not establish that personalized targeting beats the selected simple policy. A separate 873,520-row Criteo cohort tests released-sample response and selection sensitivity; it does not identify original-population causal profit.

Product analytics, experimentation, causal inference
Code and research - Evidence browser

Research portfolio

5. Operational forecasting and allocation

FreshRetailNet-LT observed sales support time-aware forecasts and capacity-constrained allocation scenarios. The implementation includes a typed ingestion ledger, immutable input snapshots and asynchronous jobs. Stockout-adjusted forecasts and operating effects remain scenarios, not observed retailer savings.

Operations analytics, forecasting, decision systems
Code and research - Evidence browser

6. Corporate Finance Decision Lab

Nine issuer reports, a traced financial spread and a funded-growth policy model. Final extraction: 33 supported correct /36 financial targets for both methods, three justified abstentions. 519 independent valuation checks and a reconciled formula workbook.

Corporate finance, FP&A, financial research and finance data
Code and research - Evidence browser

7. Indian inflation release vintages

48 official releases and 144 vintage observations. An 18-release final check retains persistence: RMSE 0.8185 pp versus 1.6901 for fixed AR(1). Release dates and the later base revision remain separate. No current inflation forecast is claimed.

Economic research, time series, publication-aware data
Code and research - Evidence browser

8. Treasury ALM and liquidity

Interest-rate hedging, published Treasury curves and hypothetical benefit cash flows. Four hedge policies, 56 instantaneous stresses and 108 liquidity-runoff cases. Independent Decimal and formula-workbook reconciliations support the modelling.

Treasury, ALM, financial risk
Code and research - Evidence browser

9. Insurance pricing and reserve uncertainty

French motor pricing and a separate US auto-reserving study. Fixed pricing comparisons and reserve interval scores expose conditional-calibration and coverage failures. The two populations are not joined.

Actuarial analytics, insurance risk, uncertainty modelling
Code and research - Evidence browser

Choose a starting point

Consumer credit / strategy: studies 1, 2, 3. Bank / model risk: studies 1, 3, 8. Insurance: studies 9, 8. Treasury: studies 8, 3. Finance / FP&A: studies 6, 7. Product / business analytics: studies 4, 5. Economic research: studies 7, 4.

Traceable data, explicit information timing, appropriate uncertainty and decisions that follow the evidence connect the studies.

Open the 9-study portfolio - GitHub - LinkedIn